Retail Sales Data Analysis Report

1. Project Overview

This project analyzes retail store sales data using Python to understand sales performance, customers, products, payment methods, and business trends.

2. Dataset Information

Original Records: 12,575
Cleaned Records: 11,362
Total Columns: 11
Missing Values After Cleaning: 0
Duplicate Rows: 0

3. Data Cleaning Summary

Converted transaction dates, handled missing values, removed rows with missing numerical values, and checked duplicate records.

4. Overall Sales Performance

Total Revenue: 1,472,998.50
Total Transactions: 11,362
Total Customers: 25
Average Transaction Value: 129.64

5. Category Performance

Butchers: 197,426.00

Electric household essentials: 192,441.50

Beverages: 187,978.50

Furniture: 186,527.00

Food: 184,645.00

Computers and electric accessories: 180,902.50

Patisserie: 172,330.50

Milk Products: 170,747.50

6. Payment Method Analysis

Cash: 513,676.00

Credit Card: 481,135.00

Digital Wallet: 478,187.50

7. Location Analysis

Online: 749,431.00

In-store: 723,567.50

8. Top 10 Best-Selling Items

Item_2_BEV: 676 units

Item_16_MILK: 627 units

Item_25_FUR: 616 units

Item_19_MILK: 589 units

Item_5_FUR: 581 units

Item_13_FOOD: 581 units

Item_1_MILK: 578 units

Item_11_MILK: 574 units

Item_11_FUR: 573 units

Item_14_FOOD: 566 units

9. Monthly Revenue Analysis

Highest Revenue Month: January
Lowest Revenue Month: October

10. Yearly Revenue Analysis

2022: 484,301.50

2023: 465,149.50

2024: 499,336.00

2025: 24,211.50

11. Final Business Insights

Best Revenue Category: Butchers
Best Payment Method: Cash
Best Performing Location: Online
Best Selling Item: Item_2_BEV
Highest Revenue Month: January
Highest Revenue Year: 2024